Comprehensive Campus Commuter Mobility and Transportation Survey 2024

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COMPREHENSIVE CAMPUS COMMUTER MOBILITY AND TRANSPORTATION SURVEY 2024

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Supervising Department: Department of Artificial Intelligence & Data Science (DEP_AIDS) in collaboration with VESIT Student Council.

Target Population: 3,000 undergraduate/postgraduate students and 200 faculty and administrative staff.

Sample Universe: 2,140 verified student respondents and 172 faculty/staff respondents.

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. SURVEY METHODOLOGY AND DEMOGRAPHIC PROFILE

In February and March 2024, the Department of Artificial Intelligence & Data Science (DEP_AIDS) deployed an exhaustive digital mobility survey across the campus community of Vivekananda Education Society's Institute of Technology (VESIT Green Campus). The objective was to construct a rigorous commuter travel model, calculate Scope 3 greenhouse gas emissions, and establish baseline indicators for institutional target TRG_GREEN_COMMUTE (70.0% low-carbon commuting share by 2027) supporting United Nations SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action).

The surveyed sample encompassed 2,312 validated respondents across all engineering departments—Computer Engineering (DEP_CMPN), Information Technology (DEP_INFT), Electronics & Telecommunication (DEP_EXTC), AI & Data Science (DEP_AIDS), and Campus Administration (DEP_ADMN).

2. COMMUTER MODAL SPLIT AND DISTANCE DYNAMICS

Cross-tabulation of daily commute journeys across 22 working days per month revealed the following modal split:

- Suburban Electric Rail (48.0% of commuter trips): Represents the backbone of student transit, utilizing the Mumbai Suburban Railway Network (Harbour Line via Chembur and Tilak Nagar stations, and Central Line via Kurla station). Average one-way distance: 14.0 km (return trip: 28.0 km).

- BEST Municipal Transit Buses (22.0% of trips): Municipal electric and CNG bus networks operated by Brihanmumbai Electric Supply and Transport (BEST), serving commuters across eastern and central Mumbai suburbs. Average one-way distance: 7.0 km (return: 14.0 km).

- Two-Wheelers (Petrol Motorcycles and Scooters) (16.0% of trips): Predominantly upper-year students and junior staff commuting from neighboring Chembur, Ghatkopar, and Sion suburbs. Average one-way distance: 6.0 km (return: 12.0 km).

- Private Passenger Motorcars (Single-Occupant and Carpools) (6.0% of trips): Predominantly senior faculty, administrative directors, and carpooling student cohorts. Average one-way distance: 9.0 km (return: 18.0 km).

- Non-Motorized Transport (Walking and Cycling) (8.0% of trips): Local students and staff residing within a 2.0 km radius of the Chembur campus. Average one-way distance: 1.0 km (return: 2.0 km).

3. TRANSPORT EMISSIONS ATTRIBUTION AND MODAL CARBON INTENSITY

Commuter mobility generates approximately 580,000 kg CO2e annually, representing 23.67% of the total institutional carbon footprint of 2,450,000 kg CO2e.

Applying verified regional emission factors:

- Private Passenger Cars: Despite representing only 6.0% of passenger trips, cars generate 165,300 kg CO2e (28.50% of transportation emissions) due to high specific carbon intensity (EF_TRANS_CAR = 0.175 kg CO2e/passenger-km).

- Two-Wheelers: Motorcycles and scooters produce 111,940 kg CO2e (19.30% of transport total) based on emission factor EF_TRANS_BIKE (0.042 kg CO2e/passenger-km).

- Municipal Buses: Produce 152,000 kg CO2e based on factor EF_TRANS_BUS (0.054 kg CO2e/passenger-km).

- Suburban Electric Trains: Account for the highest passenger-kilometres (over 5.3 million pkm) while generating only 150,760 kg CO2e due to the clean operational efficiency of electric mass transit (EF_TRANS_TRAIN = 0.028 kg CO2e/passenger-km).

- Walking and Cycling: Zero carbon emissions (0.0 kg CO2e).

4. IMPACT EVALUATION OF CAMPUS ELECTRIC SHUTTLE (INIT_ELECTRIC_SHUTTLE)

Survey data indicates that the "last-mile" transit gap between Kurla/Chembur railway stations and the campus gate historically forced over 1,200 students daily into fossil-fueled three-wheeler auto-rickshaws.

To resolve this, the college commissioned project INIT_ELECTRIC_SHUTTLE in January 2024, operating two dedicated 14-seater Mahindra Supro electric vans. In its initial 10 months of operation:

- Completed over 1,800 scheduled shuttle trips.
- Transported 38,400 passenger-trips.
- Consumed 4,464 kWh of electricity, charged on-campus using solar power from BLD_ACA.
- Displaced fossil-fuel auto-rickshaw journeys, abating 36,000 kg CO2e net.

5. STRATEGIC INTERVENTIONS TO MEET TARGET TRG_GREEN_COMMUTE

Baseline clean commuting share (train + bus + walking/cycling) stands at 52.0%. To achieve the 70.0% clean commute mandate by 2027 under TRG_GREEN_COMMUTE, the survey committee recommends:

- Procuring two 24-passenger electric mini-buses to expand INIT_ELECTRIC_SHUTTLE frequency.
- Installing 20 secure bicycle lockers and shower stations in BLD_ACA.
- Launching an AI-enabled carpool coordination module within the campus mobile portal.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_TRN_007**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*